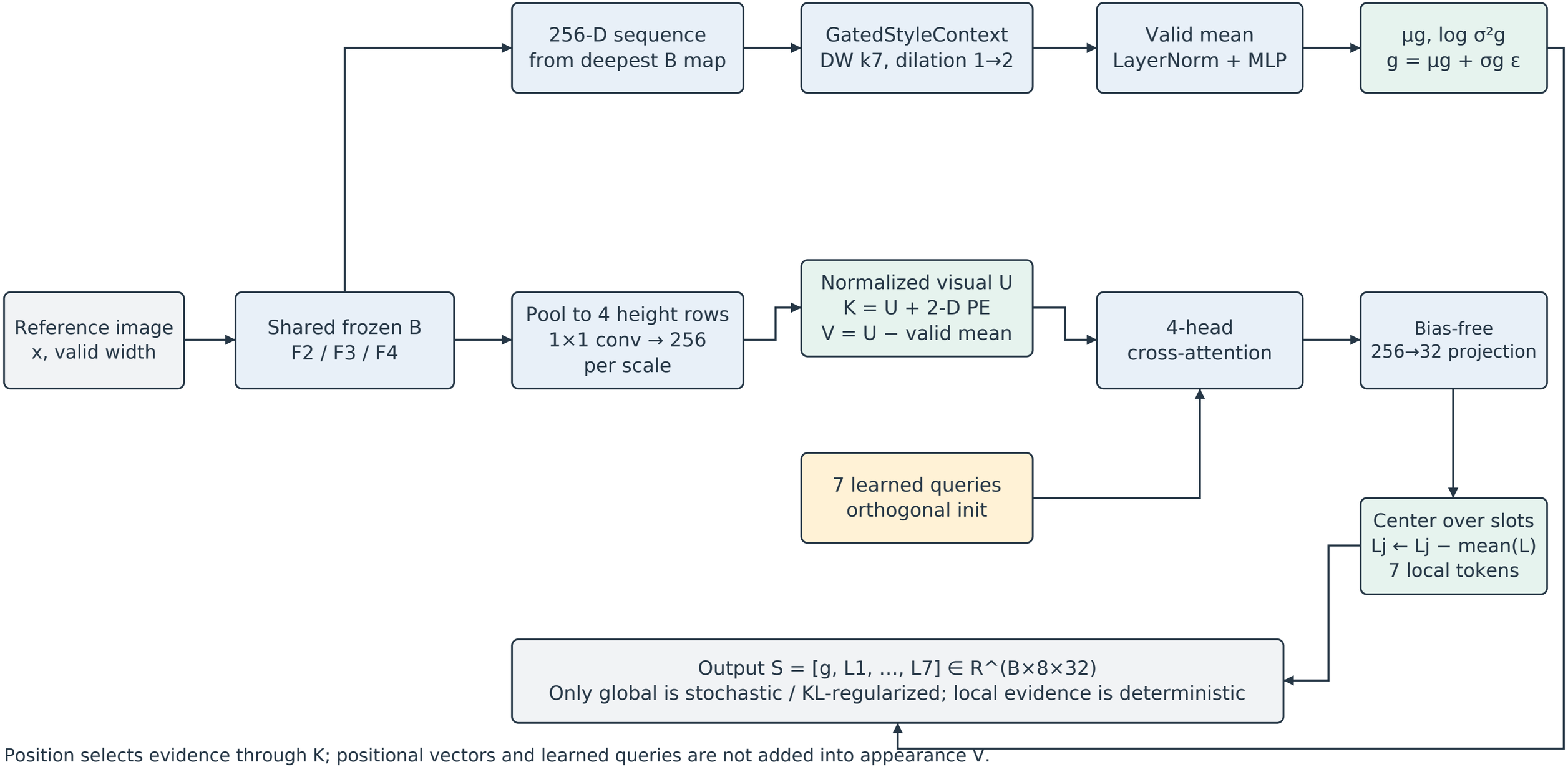
Reference image
x, valid width
Shared frozen B
F2 / F3 / F4
256-D sequence
from deepest B map
GatedStyleContext
DW k7, dilation 1→2
Valid mean
LayerNorm + MLP
μg, log σ²g
g = μg + σg ε
Pool to 4 height rows
1×1 conv → 256
per scale
Normalized visual U
K = U + 2-D PE
V = U − valid mean
4-head
cross-attention
Bias-free
256→32 projection
7 learned queries
orthogonal init
Center over slots
Lj ← Lj − mean(L)
7 local tokens
Output S = [g, L1, …, L7] ∈ R^(B×8×32)
Only global is stochastic / KL-regularized; local evidence is deterministic
Position selects evidence through K; positional vectors and learned queries are not added into appearance V.